Indispensable skills for human employees in the age of robots and AI

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Abstract

Purpose – Based on sociotechnical systems theory, social (human) and technological sub-systems in an organization should be taken in account when making strategic decisions and designed to fit the demands of the environment for organizational effectiveness. Yet there is very limited information in literature on whether employees are well equipped with indispensable (human) skills to prepare them combating challenges caused by advanced technology. The purpose of this study is to empirically investigate employees' human skills that are critical for success in the Age of Robots and Artificial Intelligence from human resource development's perspective.

Design/methodology/approach – A questionnaire was developed for the purpose of this exploratory study. A total of 422 US Midwest employees were surveyed on their human skills level that are critical for success in the Industry 4.0 transformation.

Findings – In general, the respondents could perform all the measured human skills (which can be categorized into social skillset and decision-making skillset) more than adequate but may vary by education level and gender. To strengthen one's human skills, organizations may begin with facilitating employees on relationship building to create a support system and a strong sense of belonging, which will promote their social sensitivity and collaboration skill development, as well as decision-making skillset.

Originality/value – The findings of this study can be used for techno-structural interventions and employee development programs. This study highlights the importance of investigating human skills to cope with the changing nature of work and make upskilling more feasible and flexible for workers to be robot-proof.

Keywords Sociotechnical systems theory, Indispensable skills, Industry 4.0 transformation, Techno-structural interventions, The Age of Robots and AI

Paper type Research paper

Each industrial revolution changes business structure and work pattern. The first industrial revolution involved with the development of water- and steam-powered mechanical manufacturing in 1784; the second industrial revolution began in late 19th century when electric-powered mass production based on the division of labor (assembly line) developed (Davies, 2015). In 1970s, electronics and information technology drive new levels of automation of complex tasks created the third industrial revolution. Today, many scholars and business leaders have indicated the current manufacturing is in a fourth industrial revolution, also known as Industry 4.0, because of the impacts of increased degree of technological complexity on jobs (Davies, 2015; Riminucci, 2018). For example, in summer 2022, the Procter and Gamble Co. (P&G) partnered with Microsoft to transform its digital manufacturing platform by adopting the industrial internet of things, digital twin, artificial intelligence (AI) and machine learning to reduce manual, labor-intensive tasks and improve productivities and services (Olavsrud, 2022). The increased degree of technological advancement and complexity has developed cloud-computing and self-organizing machines and changed jobs and ways people work (Wittenberg, 2016).

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The movement of technological development in each revolution not only improves productivity of organizations and employees but also changes jobs, emerges new jobs, affects workers in human–automation interaction and requires new abilities and skills to thrive in the workplace (Chuang and Graham, 2018). Analysts predicted two billion jobs worldwide will disappear by 2030 because of new scientific and technological changes (Jackson, 2014). In the USA, approximately 47% of all human jobs in manufacturing, transportation and logistics and office and administrative support will be replaced by robots, machines, automations or computerizations within the next one to two decades (Frey and Osborne, 2017). Literature has indicated a tendency of technological unemployment on middle-skill routine jobs (which comprise about 45% of works in the USA and often require precise procedures) will continually diminish in demand (Cortes *et al.*, 2017; Scopelliti, 2014). Specifically, the new computer-based technologies have:

- replaced jobs involving rules-based processing of information and repetitive physical motions;
- influenced demand for skills in the labor market; and
- sustained and created jobs comprising abstract, creative, problem-solving and coordination tasks (Autor and Dorn, 2013; Bosio and Cristini, 2018; Levy, 2010).

However, there is an in-progress demand for cognitive nonroutine skills with a requirement of college degree (known as high-skills) and manual nonroutine skills with personal traits (known as low-skills) in the US labor market (Cheremukhin, 2014; Scopelliti, 2014; Vardi, 2015). This U-shaped employment distribution, known as job polarization, can be influenced by individuals, employers, educational institutions and policymakers (Abel and Deitz, 2012). Such employment polarization – a phenomena of declined employment of middle-wage jobs and increased employment of high- and low-wage jobs – has also been observed in many European countries (Autor, 2015). Autor (2015) predicted that employment polarization will be reshaped when middle-skill jobs continually demand a mixture of tasks which combine routine and nonroutine tasks.

Although human labor survived the past two centuries of industrial revolution, the development of supercomputer, AI and robots continually creates concerns about technological unemployment (Autor, 2015). Human workers and machines have different strengths in quality production, and many studies have discussed and investigated critical issues of human–machine interaction (Autor, 2015; Bortot *et al.*, 2012; Choi and Kim, 2017; Wittenberg, 2016). Today's technological developments enable human employees moving from routine job tasks to nonroutine job tasks involving problem-solving skills, adaptability and flexibility, as well as other capabilities that human species evolved such as common sense, creativity, intuition and sensorimotor integration (Autor, 2015). In a way, employers are empowered by involving in more decision-making job tasks. Advanced technology and machines should be used to complement employees and to help them to unleash their full potential. Degani *et al.* (2017) studied human–autonomy relations based on the concept of teamwork using *object relations theory* in psychoanalytic psychology and suggested that user interactions can be evaluated by some attributes in teamwork such as sensitivity and caring. Human–machine integration can be a great social challenge. Such integration will affect the continuous development of both human and machine. According to Walmart's CEO, McMillon (2017), Walmart will be people-let and tech empowered because it is the humanity that drives creativity, competitive spirit and success. Despite its considerable investments on digital infrastructure and automation, this retail giant has put great efforts on employee training to prepare them with the latest tools and to build long-term, transferable skills for career development. In addition, the use of caretaking machines and robots may lift the burden of many women caretakers and address work hazards and economic pressures on the high cost of human caretakers in nursing homes (Parks, 2010). Yet Parks (2010) reminded us the importance of human touch by pointing out the potential destruction on social relationships caused by machines and robots. Currently, AI still lacks many of the desirable communicative aspects of human-to-human service encounters and could not handle interruptions, overlap, provide adequate communicative feedback and referential cohesion (Salomonson *et al.*, 2013).

Advanced technology is forcing the workforce moving forward. As technology developed, employees are required to upskill and reskill their competencies concurrently for gaining and maintaining employability. However, employers have reported an inadequate supply of skilled workers (who are proficient in advanced interpersonal, technical and problem-solving skills) to respond to advanced technology (Modestino, 2016). Because of the rise in nonroutine activity in automated and robotics-dependent environments, workers need to embrace technology and continually develop high-level skills and human skills, which cannot be replaced by robots, to maintain their value and stay competitive in the job market. There is a growing attention and need on developing human skills to increase the persuasive, listening, empathic and creative ability of employees.

Related literature review

Based on sociotechnical systems (STS) theory, it is vital to recognize the interaction between human behavior and society's complex infrastructures (technology and innovative work designs) in workplaces (Cummings and Worley, 2014). The STS theory offers new command and control paradigms to address challenges of modern organizational design caused by environmental complexity, dynamism, new technology and competition since the industrial revolution (Walker *et al.*, 2008). The theory suggests that a joint optimization can be achieved by a positive relationship between productivity and human fulfillment, satisfaction and commitment (Cummings and Worley, 2014). In the STS framework, both social aspect (e.g. employees' social dynamics) and technical aspect (e.g. methods, techniques and technologies) are two interacting sub-systems in an organization which should be taken into account when making strategic decisions. The social (human) aspects involve supervisory–subordinate relationships, motivation, work culture, occupational role and group membership; however, specific technical dimensions include tools, procedures, techniques, etc., that are used to accomplish job tasks (Trist and Bamforth, 1951). Each subsystem is independent from and not dominating the other. Under this notion, "an organization will function optimally only if the social and technological systems of the organization are designed to fit the demands of each other and the environment," said Pasmore *et al.* (1982, p. 1182). The theory helps managers to ensure an effective outcome of technical activities and human performances in the workplace for organizational development. Berg *et al.* (2005) highlighted that "the development and use of technology emanates from and remains linked with human values, dreams, needs and circumstances prevalent in society" (p. 346). The results of the present research of STS theory advocate the importance of human aspect in technology innovation at work.

Critical human skills

As the society has progressed to Industry 4.0, technology innovation is making some workers redundant (Frey and Osborne, 2017). Robots are now capable of performing manual tasks in manufacturing, construction, agriculture and service tasks (e.g. vacuuming, mopping and lawn mowing). Examples of irreplaceable jobs are those involving emotional and relational work, creativity, problem-solving, synthesizing and intelligent interpretations (Tripathi, 2016).

Perception and manipulation tasks (i.e. abilities to identify objects and their properties in a cluttered field of view), creative intelligence tasks (i.e. abilities to come up with novel and valuable ideas such as musical compositions and scientific theories) and social intelligence tasks (i.e. abilities to negotiate, persuade and care) are unlikely to be automated in the next decades (Frey and Osborne, 2017). Therefore, it can be argued that jobs involving human creativity and social intelligence may be protected. In view of that, there may be job skills that cannot be replaced by technology and, thus, immune from technological replacement.

Humans are unique from robots, automations and computerization because they can do well on inspiration, creativity and intuition, meaning-making, storytelling and communication (Tripathi, 2016). Human consciousness possesses sensation, thought, emotion and intuition to extract information for internalized understanding. Only human can understand and work better with human. An ability to solve unexpected challenges and problems in creative and effective ways independently and an ability to use basic communication skills (listening carefully, asking question properly and expressing opinion correctly) to analyze problems, create better plans and make better decisions with clarity and confidence are critical when working with human.

Technological change has reshaping the workplace where several foundational human skills have become essential, such as creativity, physical flexibility, sensorimotor skills, intuition, judgment, problem-solving, relationship building, collaboration, social sensitivity/intelligence, advanced communication, empathy, sense-making (commonsense), computational/critical thinking, novel and adaptive thinking, cultural intelligence and emotional intelligence (Autor, 2015; Choi and Kim, 2017; Frey and Osborne, 2017; Gajewska, 2014; Levy, 2010). Among these essential skills, creativity is widely recognized as a crucial 21st-century skill (Autor, 2015; Mansor *et al.*, 2016; Puccio, 2017). It is defined as "the generation of novel and useful ideas, whereas innovation concerns also the implementation of those ideas in the form of new practices, processes, products, or services" (Birdi *et al.*, 2012, p. 316). It is an ability to develop different ideas and new knowledge to solve new and complex problems. It is an innate survival and success skill, which can be trained and increased with management support (Boulocher-Passet *et al.*, 2016; Puccio, 2017). Creativity is related to and complements innovation; however, limited research regarding innovation process in education has been conducted (Mansor *et al.*, 2016). Because the ability to use creative thinking to solve problems is a work/life skill and can sometimes contribute to the overall organizational operation and performance in the 21st century, research in such human skill training is vital for educational practices (Mansor *et al.*, 2016; Puccio, 2017).

Sensorimotor is a critical skill in extreme situation and an important operation that human perform in daily life (Rao *et al.*, 2018). It can be trained and improved through practice in performing sensory-guided motor behaviors (Rao *et al.*, 2018). However, technology has changed the profile of sensorimotor skills and abilities such as the birth of video gaming and the death of handwriting (Heuer, 2016). Specifically, the quality of arm-hand motor control in figural movements has declined; however, the control of attention and visual orientation has been enhanced (Heuer, 2016). Karpati *et al.* (2016) investigated how intensive practice of sensorimotor skills affecting behavioral performance across various domains and discovered a significant relationship between long-term intensive dance and music trainings and sensorimotor skill enhancements. Sensorimotor is an essential skill to perform occupational activities.

All the human skills discussed above are often referred to as soft skills in literature. Soft skills are the skill of future (Tarallo, 2019). From psychological perspective, every individual has different personality traits and abilities to fit for a certain job task that often expressed in his or her soft skills, which can be developed (Ahmed *et al.*, 2013).

Human skill training and research

The 21st-century workforce requires employees to equip with technical skills and soft skills for the highly competitive global work environment (Dean and East, 2019; Griffith and Hoppner, 2013). Employers have demanded soft skills (e.g. advanced communication, teamwork and socio-emotional skills) in their employees for success in the workplace (Cunningham and Villaseñor, 2016; Groh *et al.*, 2016). Some important soft skills include communications, teamwork, motivation, problem-solving, enthusiasm and trust (Ellis *et al.*, 2014; Rasul *et al.*, 2013). Yet Dean and East (2019) discovered that oral communication, problem-solving, low self-confidence and interpersonal skills tend to be insufficient among employees. Strategic soft skill training plays a major role on effective employee and business performance outcomes (Dean and East, 2019).

Soft skill training and research can be found in all sectors across countries such as Jordan, Russia and Latin America (Groh *et al.*, 2016; Salakhatdinova and Palei, 2015; World Bank, 2010). Bailly and Léne (2013) interviewed managers in the service sector, including large-scale retailing and hotel and catering services in France, and concluded that soft skills (such as relational skills, autonomy and personification) are learnable but tend to be overlooked. Bailly and Léne (2013) additionally pointed out the importance of managers' attitudes and viewpoint toward soft skills on shaping their practices on employees' progress within the organization. Another well-documented example in Malaysia, Ibrahim *et al.* (2017) conducted a quantitative research to measure the effect of soft skills training program on the competencies of 260 managers, executives and supervisors from nine private companies. The study revealed a significant evidence of employee performance improvement via soft skill acquisition and proper training methodology adoption. Based on the findings, time-spaced learning method (i.e. giving the trainees breaks to practice and internalize what they have learned during the training program) enables trainees to transfer knowledge, obtain soft skills more effectively and increase their work performance. This study reported an important relationship between soft skills training methodology and work performance.

In organizational settings, creativity training has also been implemented but often lacks of systematic evaluation on employee innovativeness (Birdi *et al.*, 2012). Birdi *et al.* (2012) conducted an organizational field study at the UK to evaluate impact of problem-solving-based creativity training program in an international engineering firm. The study discovered short-term improvements in creative problem-solving skills and motivation to innovate for long-term improvement.

In higher education settings, training programs on individual creativity and creative teamwork and problem-solving are valued and highlighted (Salakhatdinova and Palei, 2015). Specifically, divergent thinking (open-mindedness), motivation, tolerance to risks and personal features are emphasized and assessed in the training programs for unleashing and enhancing one's creativity (Salakhatdinova and Palei, 2015). While advanced technology is changing career opportunities, interpersonal skills remain indispensable in the workplace (Fitzgerald, 2020). To keep up with the changes, educational focus is shifting from college-readiness to career-readiness, including skills like collaboration, communication, problem-solving and resilience in curriculum (Fitzgerald, 2020).

Clearly, human skills are in high demand and critical in today's workplace. Human skill training has been commonly offered in industry settings but generally lack of systematic evaluation. Hence, there is very limited information about whether employees are well equipped with human skills that prepare them to respond effectively to challenges of advanced technology.

Purpose and significance of the study

Industry 4.0 manufacturing paradigm requires advanced human skills to cope with technology sufficiently such as human–machine interaction in cyber-physical systems (Wittenberg, 2016). It transforms the traditional training to an advanced set of methods for skill development. AI and robots are taking over many routine jobs, and human employees are more engaged in creative and innovative jobs using human skills. Human skills (e.g. sensorimotor, intuition, critical analysis, judgment, collaboration, creativity, social sensitivity and problem-solving) are essential 21st-century skills for all employees to cope with the Industry 4.0 transformation. Building a more skilled, educated workforce is one of critical business strategies in its e-commerce infrastructure.

Human's ability to collect information and transmit the knowledge to create tools makes us different from other animals. Human's capability to create technology makes us unique and able to changes the world. Human employees need to advance their human (interpersonal) skills and innovative skills to set them apart from machine and robot. Technology has changed and reshaped the modern workplace. Based on the STS theory, to effectively redesign work related to the organization's technology, it is critical for human resource development (HRD) practitioners to fully understand the fundamental human skills of their employees. It is important to know what to expect in future workforce, what jobs will be replaced and, most importantly, what skills will persist and grow. Exceptional human skills do not come naturally. It is important to upgrade, up-skill and re-skill employees to prepare them to coexist with robots in the tech-driven workplaces to assist their transition to the skill-polarized labor market. The demand of human skills has continually risen, yet limited research has investigated the human skills of employees, particularly from HRD perspective. In addition, although soft (human) skill training can significantly increase participants' confidence and positivity, little evidence of its impacts on employment could be found (Groh *et al.*, 2016). "Hard skills will get you the interview – soft skills will get you the job" (Tarallo, 2019, paras.1). There is a need to further acknowledge and investigate the significant role of human skills on employees' job performance (Ahmed *et al.*, 2013) for quality life and career development.

Therefore, the purpose of this study is to empirically investigate employees' human skills that are critical for success in the Age of Robots and AI from the HRD perspective. Specifically, this study intends to identify:

- the current human skill level of employees;
- variables associated with their human skill level; and
- strategies to re-skill the employees to be robot-proof.

The research findings are expected to contribute HRD literature, as well as to assist organizational leaders and HRD practitioners to enhance employees' competitiveness in job market. Analysis of the data offers distinct information to address issues regarding upskilling the workforce and to identify potential development gap for future research. Managers and HR practitioners may use the information to maintain equivalent knowledge and to overcome limited training resources, especially for community assessment and development.

Research method

A questionnaire was developed for the purpose of this exploratory study. While there are no universal defined human skills, nine mostly recognized and commonly discussed human skills in the literature (i.e. *problem-solving*, *judgment, intuition*, *collaboration*, *relationship building*, *creativity*, *social sensitivity*, *physical flexibility* and *sensorimotor*) were comprised in the questionnaire using a five-point Likert scale: 1 = poor, 2 = less than adequate, 3 = adequate, 4 = more than adequate and 5 = excellent. The demographic questions (i.e. age, gender, education and occupation) were developed to better understand the respondents, discover meaningful insight and draw more accurate conclusion. Respondents' confidentiality was ensured, and no subjects can be identified by any means. The subjects had choices of not to answer any questions which they did not feel comfortable.

The target population was adult employees who were at least 21 years old with working experiences (either part-time or full-time) in the Midwestern USA, where its job market in manufacturing may be shrunk by advanced technology, such as electric vehicles (EVs) (Waters, 2022). Convenience sampling method was used to collect data from annual local family-friendly events at a rural area in Southern Indiana, the USA. This method is appropriate for an exploratory research at the early stage and when the population was too large (Stratton, 2021). It provided the opportunity to gather large sample size and useful data that may be challenged when using probability sampling techniques, which require more formal access to populations. To avoid potential sampling bias, all people who were present at the time and place during the data collection were asked to participate in this study. Based on Stratton's (2021) methods to improve dependability of convenience sampling, the author made sure to avoid complex study objectives, recruit as many participants as possible, describe the demographics of participants, use appropriate descriptive analysis methods, avoid over-standing findings of the study and check for possible external bias.

Based on Dillman's (2000) pretest process for content validity assurance, the questionnaire was first reviewed by three subject-matter-experts (SMEs), who were specialized in research survey development, HRD and employee training, for clarity on questions and congruency on structure. The instrument was revised according to the SMEs' feedback and then reviewed by ten adult workers from the target population for readability, impression and completing time of the instrument before the distribution. The instrument was again revised based on the respondents' feedback and ready for distribution.

A screening device was developed in the questionnaire to ensure no participant filled out the questionnaire more than once. Additional questions regarding the respondent's residency and whether it was the first time to fill out this questionnaire were added.

Findings and discussion

In all, 422 responses were collected and used for data analyses. Genders were closely equal represented in the data set (men n = 196, 46.9%; women n = 222, 53.1%). The majority of respondents were white (n = 353; 84.0%), which is similar to the population of the rural area where about 80% of the residents were white. No subgroup was under-representation or over-representation, as the demographic data could be broken down properly. In terms of age (M = 37; range = 18–76 years old; SD = 13.8), about 65% of the respondents are members of Generation Y (born in 1980–1994; n = 146; 34.8%) and X (born in 1965–1979; n = 128; 30.5%). Most of the respondents (n = 367; 87.4%) held either a high school (n = 180; 42.9%) or a bachelor (n = 187; 44.5%) degree. About 76% (n = 319; 75.6%) of respondents were working full-time, while the rest of respondents were either part-time (n = 98; 23.2%) or contractor (n = 5; 1.2%). The most common occupations held by the respondents were: management (n = 67; 15.9%); health care (n = 43; 10.2%); education, training and library (n = 37; 8.8%); sales (n = 33; 7.8%); food preparation and serving (n = 32; 7.6%).

In general, the respondents reported they could perform all of the human skills more than adequate (n = 422; M = 4.09; SD = 0.57; Table 1). To examine the nine human skills, each specific skill was rated more than adequate, except *physical flexibility*, such as joint range of motion and muscle extension (n = 422; M = 3.78; SD = 0.91). *Physical flexibility* was not expected to be rated as the least adequate skill because about 65% of the respondents were the Generation X and Y cohorts which were in their young or middle adulthood. This finding raised a question – whether age played a role in this result. The finding may be explained by the fact that three in four adults in the USA do not get enough physical activities to increase their physical flexibility, coordination and strength (Centers for Disease Control and Prevention, 2022). Proper physical activity can improve wellness and help people function better throughout the day (Centers for Disease Control and Prevention, 2022).

Correlation between demographics and human skills

Pearson correlation coefficient analysis was conducted to measure the relationship between demographics and the overall human skill level. Calculated correlations were analyzed using Davis's (1971) conventions for describing measures of association (0.00–0.09 = negligible, 0.10–0.29 = low, 0.30–0.49 = moderate, 0.50–0.69 = substantial, 0.70 and higher = very strong and 1.00 = perfect). Age and gender factors showed no influence on the respondents' human skill level (Table 2). To answer the question raised from the last section, age did not play a role in *physical flexibility* being rated the least adequate skill. Yet a low level of correlation (r = 0.108) between education and human skill level was found.

Table 1. Descriptive statistics of respondents' overall performance level of the human skills (N = 422)

	n	Minimum	Maximum	M	SD
Human skills	422	2.4	5	4.09	0.57
• Judgment	420	1	5	4.23	0.73
• Problem-solving	421	1	5	4.22	0.77
• Relationship building	422	1	5	4.15	0.86
• Sensorimotor	417	1	5	4.12	0.78
• Collaboration	422	1	5	4.11	0.83
• Intuition	422	2	5	4.11	0.75
• Social sensitivity	421	1	5	4.04	0.97
• Creativity	420	1	5	4.04	0.90
• Physical flexibility	422	1	5	3.78	0.91

Notes: Response scale: 1 = poor, 2 = less than adequate, 3 = adequate, 4 = more than adequate and 5 = excellent

Table 2. Correlation analysis between demographics and human skills (N = 422)

	n	M	SD	Age	Gender	Education
Age[a]	419	2.46	0.99			
Gender[b]	418	1.53	0.50	−0.02		
Education[c]	420	3.22	1.17	0.07	0.05	
Human Skills	422	4.09	0.57	0.07	0.00	0.11*

Notes: *$p < 0.05$. [a]coded 1 = Generation Z, 2 = Generation Y, 3 = Generation X, 4 = Baby Boomer and 5 = Traditionalist. [b]coded 1 = Male and 2 = Female. [c]coded 1 = Others, 2 = High School, 3 = Associate, 4 = Bachelor, 5 = Master and 6 = Doctoral

Because education levels were not equally represented among the respondents, additional correlation coefficient analysis was conducted to further examine the identified relationship between education and human skills. As presented in Table 3, the results indicated significant (low level of) relationships between education and *intuition* (r = 0.140) and *relationship building* (r = 0.109). Each skill was significantly related to another skill. *Physical flexibility* appeared to have either low or moderate associations with other skills, while other skills tended to have higher correlations with others. A strong correlation was identified between *judgment* and *intuition* (r = 0.70). Based on the measured associations, the human skills could be categorized into social skillset and decision-making skillset (Figure 1). As illustrated in the figure, *social sensitivity*, *collaboration* and *relationship building* are strongly correlated. Together, this social skillset triangle may increase one's *problem-solving* and *judgment* skills, which may lead to one's *intuition*, *creativity*, *sensorimotor* and *physical flexibility* improvement. This figure suggests that, to strengthen one's human skills, organizations can begin with facilitating employees on relationship building for support system and a strong sense of belonging, which will then enhance their *social sensitivity* and *collaboration* skills, as well as decision-making skillset. Based on Lewin's *group dynamics theory*, it is possible that the ability to cooperate and to induce cooperation in others create open and complex systems in group that affect the behavior of the group (Gençer, 2019).

The decision-making skillet finding in Figure 1 can be supported by a number of early research in literature. For example, intuition has been viewed as a highly adaptive decision-making tool which may lead to better outcomes than deliberative thinking (Gigerenzer and Gaissmaier, 2011; Lieberman, 2000). Intuition can be accurate in certain kinds of judgments (Gigerenzer and Gaissmaier, 2011).

A statistically significant difference among the education groups was found on the *relationship building* (Table 4). Respondents with master's degree rated the highest on *relationship building* which was significantly different from other groups, followed by respondents with doctoral degree and bachelor's degree. The perceptions of human skill level on *relationship building* could be varied based on individuals' education backgrounds. The finding suggests that the higher level people are educated, the greater values they have on *relationship building*. The 20th-century education policy on aspect of relationship building and teamwork (Delors, 1998) may play a significant role in this finding. As André Gorz (1988) stated in book "Métamorphoses du travail: Quête du sens":

> [. . .] education must reverse its priorities; instead of training 'human computers' whose capacities for memorizing, analyzing and calculating are exceeded and largely made redundant by computers, we should concentrate on developing distinctively human skills: manual, artistic, affective, interpersonal and moral skills and the ability to ask unexpected questions, to make sense of things and to reject nonsense even when it is logically coherent (Bertrand, 1998, p. 165).

Gender differences in relationship building and physical flexibility

No significant results were found in Levene's test for equality of variances in all variables. The findings from *t*-tests indicated no significant differences between genders on the overall human skill level (Table 5). However, statistically significant differences between men and women were found on *relationship building* ($t = -2.65$; $p = < 0.05$) and *physical flexibility* ($t = 2.21$; $p = < 0.05$).

The finding of women rated significantly higher on *relationship building* may be because that women are known for their empathy and social skills, competing on being nice and liked and seeking security in friendship (Mandell and Pherwani, 2003). Literature has found that women are comprehensive processors who tend to seek association between their self-

Table 3.
Correlations between education and nine human skills (N = 422)

	n	M	SD	Education[a]	Sensorimotor	Physical flexibility	Intuition	Judgment	Creativity	Problem-solving	Relationship building	Collaboration
Education	420	3.22	1.17									
Sensorimotor	417	4.12	0.78	0.06								
Physical flexibility	422	3.78	0.91	0.06	0.44**							
Intuition	422	4.11	0.75	0.14**	0.50**	0.37**						
Judgment	420	4.23	0.73	0.07	0.42**	0.38**	0.70**					
Creativity	420	4.04	0.90	0.07	0.19**	0.25**	0.36**	0.41**				
Problem-solving	421	4.22	0.77	0.09	0.43**	0.29**	0.47**	0.58**	0.42**			
Relationship building	422	4.15	0.86	0.11*	0.34**	0.25**	0.35**	0.40**	0.35**	0.41**		
Collaboration	422	4.11	0.83	0.03	0.38**	0.29**	0.42**	0.43**	0.33**	0.49**	0.63**	
Social sensitivity	421	4.04	0.97	0.04	0.32**	0.28**	0.30**	0.36**	0.28**	0.32**	0.65**	0.58**

Notes: $*p < 0.05$; and $**p < 0.01$. [a]coded 1 = Others, 2 = High School, 3 = Associate, 4 = Bachelor, 5 = Master and 6 = Doctoral

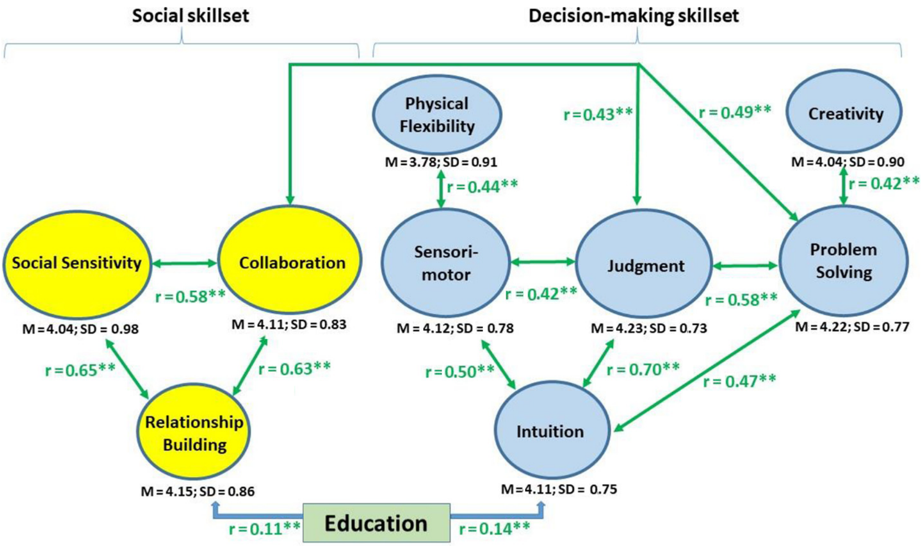

Figure 1. Correlations between Education and Human Skillsets

Table 4. ANOVA analysis: perceived human skill level of employees by education (N = 422)

	Doctoral (n = 7)		Master (n = 37)		Bachelor (n = 187)		Associate (n = 4)		High School (n = 180)		Others (n = 5)		
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD	F[a]
Intuition	4.14	0.90	4.38	0.59	4.16	0.71	4.25	0.50	4.01	0.80	3.80	0.84	2.03
Relationship building	4.29	0.95	4.54	0.65	4.15	0.87	4.00	1.41	4.11	0.85	3.40	1.14	2.45*

Notes: $*p < 0.05$. [a]One-way ANOVA (df = 5, 414)

generated and outside information (Bhaduri and Ha-Brookshire, 2015). A more recent study reported that women leaders are more likely to use instructional leadership approach with a focus on relationships than men (Shaked *et al.*, 2019). Evidently, women tend to value higher on human relations as an important life skill. In addition, it is interesting to discover that both education and gender play an influential role on *relationship building*.

Women respondents reported less adequate on *physical flexibility*. The result is consistent with a common understanding about gender differences in human physiology. In general and under the same condition, men are physically stronger than women because of the higher muscle-mass to body-mass ratio (Maughan *et al.*, 1983). However, in terms of human capabilities, machines and robots cannot completely exceed human physiological characteristics although AI may overtake human intelligence such as AlphaGo, the first self-learning program to defeat a world champion in the game of Go (Silver *et al.*, 2017).

Implications

This study has several implications for HRD and STS research. Based on a theoretical lens of STS theory, this study advances the HRD literature by empirically testing human skills that are critical for workers in the Age of Robots and AI. Human skills are unique capabilities of human being which are critical job skills in responding to the dynamic of

Table 5.
Mean differences between genders on human skills

	Male			Female			
	n	M^a	SD	n	M^a	SD	t
Human skills	196	4.09	0.58	222	4.09	0.56	−0.01
• Problem-solving	195	4.29	0.76	222	4.17	0.79	1.53
• Judgment	196	4.24	0.72	221	4.21	0.73	0.51
• Sensorimotor	192	4.16	0.77	221	4.09	0.79	0.85
• Intuition	196	4.13	0.74	222	4.08	0.77	0.76
• Collaborating	196	4.07	0.87	222	4.16	0.81	−1.16
• Relationship building	196	4.04	0.92	222	4.26	0.80	−2.65*
• Creativity	195	4.03	0.97	222	4.04	0.84	−0.17
• Social sensitivity	196	3.97	1.00	221	4.11	0.95	−1.50
• Physical flexibility	196	3.89	0.96	222	3.69	0.85	2.21*

Notes: [a]Response scale: 1 = poor, 2 = less than adequate, 3 = adequate, 4 = more than adequate and 5 = excellent. $*p < 0.05$

destructive social relationships caused by robots and machines (Ahmed *et al.*, 2013; Parks, 2010). The findings offer a fundamental work of integrating the STS framework in employee training and development. Moreover, because the STS theory has often been adopted in areas outside HRD, such as hospitality management (Lee, 2022), information technology (Berg *et al.*, 2005), computer sciences (Trabert *et al.*, 2022), the findings of significant relationships between human skills, gender and age contribute to a better understanding of human skill development that drive performance in the tech-driven workplaces. In addition, this study addresses a current literature gap on employees' readiness with indispensable (human) skills to combat challenges caused by advanced technology from HRD perspective. An implication for gender scholarship is that the gender effect could cause different needs on relationship building and physical flexibility between men and women which may influence their competitiveness on the tech-savvy job market.

This study also has several practical implications for HRD professionals. The recent attention to the green energy and limited nature resources has pressed the development of EVs. With AI, automation and new design of EVs, the modern factories would only need a smaller number of workers than they were in the past when producing gas cars in assembly lines. There were nearly 20 million people worked in manufactory around 1980, but the number is sown to close to 13 million in 2019 (Harris, 2020). Assembly workers, for example, are competing in a smaller pool of jobs with a need for advanced training to upskill and reskill for the next job. Along with the job market and work environment changed, they will be transferred to jobs which require another set of skills. This study confirmed that human employees could generally perform all of the human skills more than adequate. To facilitate employees' continuous development for today's tech-driven workplaces, HRD professionals should begin with understanding the critical effect of human aspect on digital infrastructure and automation and offer supports to enhance employees' human skills and advanced technical skills required by new jobs.

Second, the findings on significant relationship among employees' human skill levels provide important guidance in employee development programs for HRD practitioners. To reskill and upskill employees to be robot-proof, human skills can be enhanced based on social and decision-making skillsets. Because each human skill is significantly related to another, the social skillset triangle can be a good starting point to improve individual human skills. As the findings indicated, organizations can begin with facilitating employees on relationship building to create a support system and a strong sense of belonging, which will then support their social sensitivity and collaboration skill development, as well as decision-making skillset.

Another practical implication of this study is to use the findings for techno-structural interventions (such as competence-based management and job design/enrichment) and employee development programs (such as career development and training) via an improvement-based approach to get a best fit to the current situation – the Age of AI and the Age of Remote Work. The automation has challenged the productivity of human labor. Along with the impact of COVID-19 pandemic, an issue of *quiet quitting*, which employees tend to do only minimum required job rather than going above and beyond, has been creating another challenge to the workforce development in the EV manufacturing. According to Gallup finds, at least 50% of US workers are quiet quitters since second half of 2021, resulting in low engagement and job resignation, and the job disengagement and dissatisfaction were rated higher by remote younger millennials and Generation Z (Harter, 2022). Yet AI and robots cannot lead a team, build a culture, develop a trusted relationship, etc. To address the issues, HRD professionals could increase their efforts on talent recruitment and retention through techno-structural interventions and industry-specific training in human skills. In light of STS theory, social and technological systems of the organization should be designed to fit the demands of the environment. To improve employee's performance and organizational effectiveness during the environmental changes, carefully planned approaches to work redesign, talent development, virtual learning environment and well-being in the workplace that focus on humanization are critical.

Conclusion and recommendations for future research

Robots and AI technologies are developed to enhance human job performance and make human lives better and easier. However, the challenges of technology unemployment and job change have required continued HRD efforts on employees' reskilling and upskilling for organizational effectiveness. As Vittorio Cretella, the CIO of P&G, suggested, organizations need to invest in people to drive organizational digital transformation (Olavsrud, 2022). This study examined employees' human skills that are critical for success in the Age of Robots and AI. The results highlight the importance of investigating human skills at multiple levels of analysis and offer approaches to enhance employees' human skills to thrive in the future digital workplace. It is important for HRD practitioners to nurture employees' human skills, which machines cannot really master, for the changing nature of work and make upskilling more feasible and flexible for workers. Strategies to re-skill the employees to be robot-proof are crucial for continuous performance improvement at all levels in organizations.

The results of this study serve as a foundation for future research to investigate positive impact of education on values of human skill such as relationship building for technological future readiness. Given that *physical flexibility* was rated the least adequate skill and had low correlations with other skills, future research regarding the influence of *physical flexibility* on sensorimotor and decision-making from HRD perspective would develop a better understanding on critical issues of human capabilities in the Age of AI. Additionally, because this study used a non-probability sampling method, the findings cannot be generalized to the population as a whole. Convenience sampling was used to develop objectives for more rigorous research (Stratton, 2021). Replication using probability sampling is recommended to see if the findings are replicable and for more reliable results. Moreover, in a STS, human factor has a strong impact on value creation and business model development (Trabert *et al.*, 2022). During times of change such as Industry 4.0 transformation and the COVID-19 pandemic, the value of techno-structural interventions and employee development programs to redefine the new meaning of work and advance human skills in the virtual work environment could be investigated. Finally, in an exploratory phase, limited research works on importance of humanization in today's workplace were found. Future research requires more detailed information concerning interaction effects between organizational technical aspect and social (human) aspect in the Age of AI and the Age of Remote Work.

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